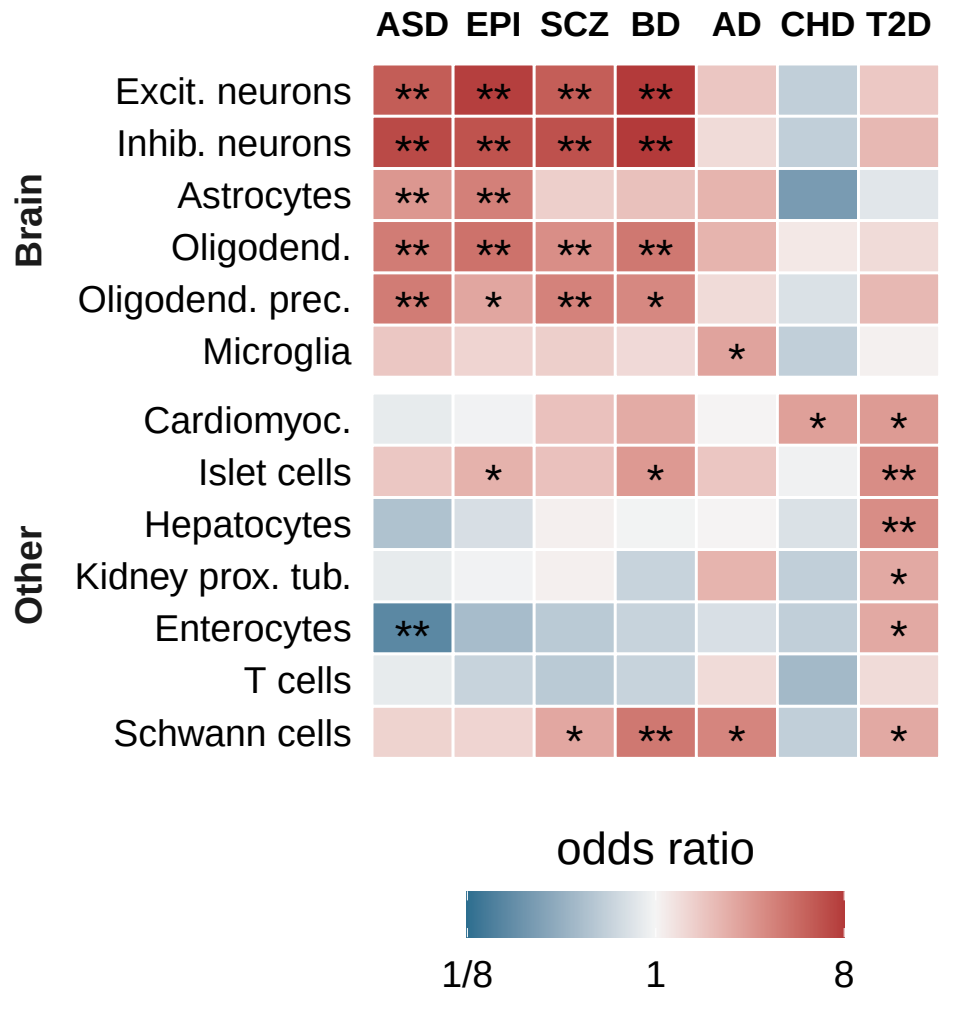
a

ASD
EPI
SCZ
BD
AD
CHD
T2D
p = 1.6 × 10−20
p = 3.2 × 10−13
p = 0.011
p = 1.1 × 10−4
1
10
100
OR (>= Existing)
in lowest LOEUF decile

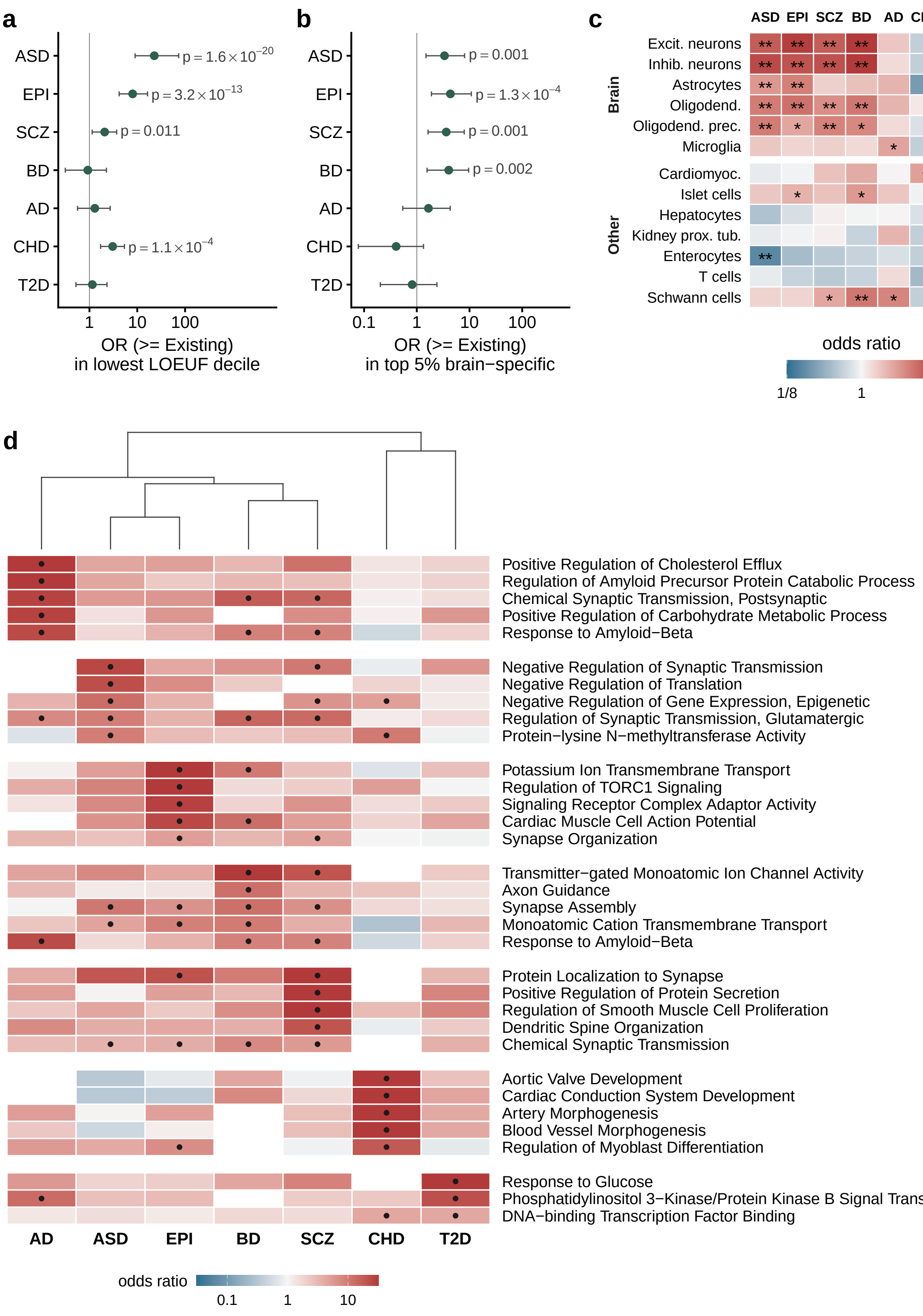
b

ASD
EPI
SCZ
BD
AD
CHD
T2D
p = 0.001
p = 1.3 × 10−4
p = 0.001
p = 0.002
0.1
1
10
100
OR (>= Existing)
in top 5% brain−specific

c


ASD EPI SCZ BD AD CHD T2D
Brain
Excit. neurons
Inhib. neurons
Astrocytes
Oligodend.
Oligodend. prec.
Microglia
Other
Cardiomyoc.
Islet cells
Hepatocytes
Kidney prox. tub.
Enterocytes
T cells
Schwann cells
odds ratio
1/8
1
8

d


Positive Regulation of Cholesterol Efflux
Regulation of Amyloid Precursor Protein Catabolic Process
Chemical Synaptic Transmission, Postsynaptic
Positive Regulation of Carbohydrate Metabolic Process
Response to Amyloid−Beta
Negative Regulation of Synaptic Transmission
Negative Regulation of Translation
Negative Regulation of Gene Expression, Epigenetic
Regulation of Synaptic Transmission, Glutamatergic
Protein−lysine N−methyltransferase Activity
Potassium Ion Transmembrane Transport
Regulation of TORC1 Signaling
Signaling Receptor Complex Adaptor Activity
Cardiac Muscle Cell Action Potential
Synapse Organization
Transmitter−gated Monoatomic Ion Channel Activity
Axon Guidance
Synapse Assembly
Monoatomic Cation Transmembrane Transport
Response to Amyloid−Beta
Protein Localization to Synapse
Positive Regulation of Protein Secretion
Regulation of Smooth Muscle Cell Proliferation
Dendritic Spine Organization
Chemical Synaptic Transmission
Aortic Valve Development
Cardiac Conduction System Development
Artery Morphogenesis
Blood Vessel Morphogenesis
Regulation of Myoblast Differentiation
Response to Glucose
Phosphatidylinositol 3−Kinase/Protein Kinase B Signal Transduction
DNA−binding Transcription Factor Binding
AD
ASD
EPI
BD
SCZ
CHD
T2D
odds ratio
0.1
1
10